

Breast Cancer Detection: Wisconsin Diagnostic Dataset

ISA 640 – Capstone Project

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Recap:

Previously

Quick Updates:

Data sets & industry analysis:

- Clinical + pathology
- Genomics (genes analysis)
- CNN or computer vision on imaging
 - Ultra-sounds
 - Mammograms
 - I. Detection
 - II. Classification
 - III. Survival Analysis

Features Engineering:

- 27 features: irregularities in (size & shape)
 - Radius, perimeter, area, symmetry, direction

Quick Updates:



Today:

Baseline Model (LR):

- Untuned with 5-fold cross validation
- Tuned with GridSearch
- Evaluation (precision, recall, F!, ROC, Confusion matrix)

Alternative Models:

- KNN
- SVM
- Decision Tree
- RF
- GBT
- Ensemble Learning (voting, bagging, boosting, stacking)

Thank you!

Questions?

